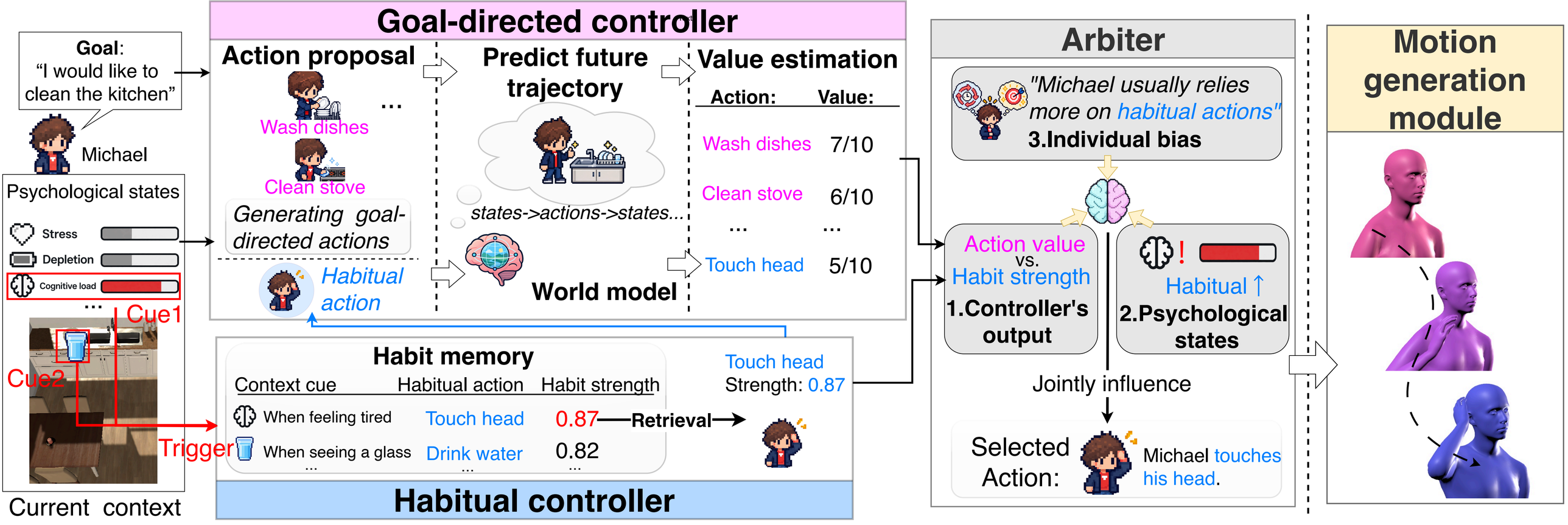

Goal-directed controller
Goal:
"I would like to clean the kitchen"
Michael
Psychological states
Stress
Depletion
Cognitive load
...
Cue1
Cue2
Trigger
Current context
Action proposal
Wash dishes
...
Clean stove
Generating goal-directed actions
Habitual action
Predict future trajectory
states->actions->states...
World model
Value estimation
Action: Value:
Wash dishes 7/10
Clean stove 6/10
... ...
Touch head 5/10
Habit memory
Context cue Habitual action Habit strength
When feeling tired Touch head 0.87
When seeing a glass Drink water 0.82
Retrieval
Touch head
Strength: 0.87
Habitual controller
Arbiter
"Michael usually relies more on habitual actions"
3.Individual bias
Action value
vs.
Habit strength
1.Controller's output
Habitual ↑
2.Psychological states
Jointly influence
Selected Action:
Michael touches his head.
Motion generation module